

Capstone Checkpoint 1.1: Scoping and Initial Agent Design

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1. Selected Agent Option & Problem Framing

- **Selected Agent Option:** Personal Assistant
- **Intended User:** Individuals, home offices, or privacy-conscious professionals managing large volumes of physical mail, tax forms, utilities, and legal documents.
- **Problem Statement:** Busy people do not have time to sort physical mail, leading to cluttered counters, forgotten bills, and costly late fees and time wasting pulling up important information. Manual filing is a chore, and simple scripts break when document layouts change. This local, privacy-first system automates the pipeline entirely. Drop a mixed stack into the scanner, hit scan, and walk away. The engine reads, categorizes, extracts metadata, and archives every file into a searchable database. For optional physical tracking, the system can assign a batch number to each scanned stack. Searches return both the digital file and its physical batch location for quick retrieval from storage, though users can also bypass this entirely and rely strictly on digital copies.

2. Inefficiency of Stand-Alone LLMs and Simple Prompting

A stand-alone language model or single prompt pass is fundamentally insufficient for this problem due to state management and execution boundaries:

- **Lack of Environmental Interaction:** A single prompt cannot read binary PDF streams from disk, calculate SHA-256 byte hashes for deduplication, run local OCR on image-only scans, or physically execute filesystem moves.
- **Cascading Logic Failures:** Document sorting requires deterministic accuracy. A single LLM pass often hallucinates non-existent categories or misformats dates. Without an agentic loop, a single prompt failure leaves the document stranded without recovery.
- **Contextual Blindness:** An LLM prompted in isolation lacks memory of past human corrections or corporate aliases (e.g., mapping "PWSA" to "Pittsburgh Water and Sewer Authority").

3. Agent Environment

The agent operates within a hybrid local environment consisting of:

- **Data Sources & Ingestion Streams:** Unstructured local file directories where new digital PDFs and digitized physical scans are dropped.
- **Perceptual & Optical Tools:** Native text extraction parsing modules and optical character recognition (OCR) engines for image-only scans.

- **Persistent State & Memory Base:** A structured local database containing metadata manifests, entity mapping rules, and historical feedback logs from past user corrections.
- **Execution Environment:** The local filesystem hierarchy where classified documents are renamed, categorized, and archived or flagged for review.
- **Human Interface Layer:** An evaluation interface that prompts the user for clarification when the agent detects unresolvable ambiguity or low classification confidence. Potential for a localized self learning/patching concept.

4. Agent Actions

To progress toward its goal, the agent operates in an autonomous ReAct loop using four core tools:

- **Deduplicate:** The agent checks whether the incoming document has already been processed to avoid redundant work.
- **Inspect & Extract Content:** The agent examines the visual quality of the document and chooses the appropriate method to extract readable text (e.g., direct text parsing for digital files versus optical recognition for low-quality scans).
- **Recall Context & Classify:** The agent queries its internal memory for relevant rules, past user corrections, or document patterns, using that context to determine key metadata (such as document type, date, and source).
- **Validate & Archive:** The agent verifies that all required metadata fields are complete and accurate before applying standardized naming rules and filing the document into the correct destination path.

5. Feedback Mechanisms and Self-Correction

Feedback guides the agent across multiple operational steps:

- **Internal Programmatic Feedback:** If tool execution returns an invalid category or a malformed date string, the agent catches the validation error, adjusts its extraction strategy (e.g., expanding the context window or re-running OCR with layout preservation), and retries the classification step.
- **Human-in-the-Loop Learning Feedback:** When the agent cannot reach high confidence, it routes the document to a manual review queue. When a human corrects the record via the UI, the system logs the correction into a separate internal learning mechanism, or at least into a pattern recognition component that can be called during the feedback loops to improve subsequent scans.